

At Most Half a Run: Guarantees and Limits of Reversal-Based Detection of Order-Dependent Flaky Tests

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Submission to the ICST 2027 Research Track

Abstract—Should detectors of order-dependent (OD) flaky tests spend their rerun budget on reversed test orders? iDFlakies does, because the reverse of a passing order is more likely to fail. We show that this intuition is correct but its payoff is bounded. First, we prove that an order and its reversal have nonpositively correlated failure events in four structural classes of OD tests, which together cover all 289 published OD models we study. The proofs combine a composable sum-of-squares certificate, a cyclic-cut reduction to Harris’ inequality, and a two-square window identity; counterexamples mark where the property fails. Second, we prove a stopping rule: independent reversal pairs save at most half a suite run in expected time to a test’s first failure, and independent k -run blocks at most $(k - 1)/2$ runs when $kf \leq 1$, where f is the failure probability. Outcome-gated reversal, as deployed, changes the order distribution and inherits neither guarantee. Simulating the 289 models in 47 modules, pairing saves 0.40 runs per test and adds 0.19–0.33 percentage points at 20 runs. Real executions of two complete modules, 300 order/reversal pairs each, match all 16,200 predicted outcomes of 27 OD tests. Finally, we derive the accuracy a reversal predictor needs to beat random orders, which turns the remaining gap into a concrete target.

Index Terms—flaky tests, order-dependent tests, test order, reversal, probabilistic analysis, antithetic sampling

I. INTRODUCTION

An order-dependent (OD) flaky test can pass or fail on the same code depending on which tests run before it [1], [2]. Detectors therefore spend a limited budget executing a suite in different orders [3]–[5]. The engineering question is not just whether a new order is more likely to fail, but *whether choosing it saves enough executions to matter*.

Reversal is a particularly consequential example. Wei et al. [6] argued that reversing a passing order can move a polluter before its victim, and observed a higher conditional failure probability by sampling. iDFlakies adopted reversal when a round finds no new flaky test [6], [7]. This leaves two gaps: a conditional probability gain is not a detection-time saving, and a statement about one passing test does not justify a gate driven by the whole suite.

Our answer is a limit, not a faster detector. Independent reversal pairs save at most *half a suite run per target* in expected time to its first failure, for any fixed failure event on the sampled orders. For a target failing in 1% of random orders, even optimal pairing reduces 100 expected runs only to 99.5, not 50. For independent blocks of k uniform runs, the saving is at most $(k - 1)/2$ when $kf \leq 1$. Thus a fixed

small block cannot yield a constant-factor improvement on increasingly rare failures. This rules out a broad family of local scheduling changes as a remedy for rare OD tests; it is stronger than observing a small gain on one dataset.

What readers can use.

- *Detector builders*: distinguish unconditional pairing from a suite-level gate. Pairing is provably no worse than independent sampling under explicit OD-model hypotheses; a gate needs its own evaluation. At iDFlakies’ default of 20 runs, our simulation finds no reason to replace the deployed gate.
- *Researchers*: use the half-run and block limits to screen proposed schedules before expensive evaluation. To beat these limits, change the schedule class: coordinate longer blocks or use information about which orders will fail. We derive an exact accuracy/cost tradeoff for an imperfect reversal predictor.
- *Evaluators*: separate model validity, simulator correctness, and actual executions. A reference-simulator match establishes only the second. State the passing-reference assumption and evaluate outcome-dependent policies jointly over a module.

Evidence and scope. We prove reversal’s nonpositive failure covariance for four structural cases covering all 289 published OD models studied here. We simulate all those models and separately execute two complete modules to check matched-order predictions (Section V-G). At 20 runs, pairing adds 0.19–0.33 pp over random orders. Of the 289 models, 108 fail the recorded original order under the abstraction; the 181 reference-consistent models are reported separately. Such failures can be genuine: in an executed module, the recorded order really fails the six affected targets.

Contributions and reading path. Section IV gives three central results with their proof ideas: when pairing is safe, its half-run/block ceiling, and lower bounds for broader scheduling classes. It also separates the deployed gate from those guarantees and turns an ideal oracle into a measurable predictor requirement. Section V measures effect sizes and abstraction sensitivity; Section VI states the resulting tool decisions. The supplement contains complete proofs, the frontier conjecture, and exact checks. Practical readers can skip the proofs.

TABLE I
THE THREE-TEST EXAMPLE: CLASSES $\{v, c\}$ AND $\{p\}$.

Order σ	v fails	$R\sigma$	v fails in $R\sigma$
vcp	no	pcv	no
cvp	no	pvc	yes
pvc	yes	cvp	no
pcv	no	vcp	no

II. BACKGROUND AND MOTIVATION

OD tests. Following Wei et al. [6], a victim v with polluters P and polluter-specific cleaner sets C_p fails in order ω iff

$$\exists p \in P : p \prec_\omega v \wedge \nexists c \in C_p : p \prec_\omega c \prec_\omega v. \quad (1)$$

A brittle fails iff no state-setter precedes it. Li et al. [8] use these semantics in their simulator for 289 OD tests, and they count a test as detected once both a passing and a failing order have been observed. We report this protocol and a known-reference protocol (Section III).

The reversal gate in iDFlakies. At the commit we inspected [7], `RandomDetector` requests a fresh shuffled order unless the previous round’s filtered result (new, confirmed flaky tests) is empty. In that case it issues the reverse of the previous order. `TestShuffler` falls back to a fresh order if that reversal has already been issued, so an order is never reversed twice in a row. The source also contains a commented-out alternative that reverses only after a run without any failure. We call the deployed rule `GATE` and the alternative `STRICT`.

A three-test example. Let classes be $\{v, c\}$ and $\{p\}$, where p pollutes v and c cleans. Table I lists all orders. The four legal orders are vcp , cvp , pvc , pcv , and only pvc fails, so the flake rate is $f = 1/4$. The reverse of the passing order cvp is the failing order pvc . Thus $\mathbb{P}(\text{reverse fails} \mid \text{pass}) = 1/3 > f$, which is the effect Wei et al. exploit. Now pair a random order with its reversal. No order fails in both directions, so one pair detects v with probability $2f = 1/2$; two independent orders detect it with probability $7/16$. In expected runs until the first failure, pairing needs 3.5 runs and independent sampling needs 4. The saving is exactly half a run, and Theorem 2 shows that half a run is the most independent reversal pairing can save.¹

III. MODEL

Definition 1 (Hierarchical legal orders). A rooted tree has the tests as leaves. At every internal node, the children are permuted uniformly and independently of other nodes, and each child’s subtree runs contiguously. The induced distribution μ on the set Ω of legal orders is uniform. The map R reverses a whole order; it corresponds to reversing every node’s permutation, so R is a measure-preserving involution.

¹The closed form printed for this setting (“Special Case 2” in [6, Sec. 4.4]) gives $\mathbb{P}(\text{pass} \wedge \text{reverse fails}) = 1/2$ here instead of $1/4$. In general it overstates the probability by $\gamma_1 S_\pi / [k(\pi_1 + \gamma_1 + 1)]$ in the notation of [6]. The reported comparison in [6] was obtained by sampling. Corrected formulas and an exhaustive check are in our package.

Class-contiguous orders form the two-level case (root $\rightarrow$ classes $\rightarrow$ methods). Deleting tests that do not affect a target keeps the induced order uniform on the pruned tree, so only relevant tests matter.

Common-cleaner semantics. Fix a victim v with pairwise disjoint sets P and C . Every run starts with a clean bit. Polluters set the bit, cleaners clear it, and v fails iff the bit is set when v runs. This is Eq. (1) with $C_p = C$ for all p .

Target statistics. For a target, let $F = \{\sigma : \text{target fails in } \sigma\}$ and

$$f = \mu(F), \quad B = \mathbb{P}(F(\sigma) \wedge F(R\sigma)), \\ q = 1 - 2f + B, \quad d = f^2 - B.$$

B is a joint probability; the covariance of the two failure indicators is $-d$. Because R preserves μ , the order and its reversal both fail with probability f , and q is the probability that both pass.

Detection protocols. With a *known passing reference* (KR), a passing outcome is supplied externally; this is an assumption, not a model-consistency claim about the recorded original order, and a target is detected at its first failing run. With *no reference* (NR), the definition of Li et al. [8], both a failing and a passing run must occur within the budget.

Schedules. A budget of r complete suite runs is spent by IID (fresh uniform orders), PAIR ($\sigma_1, R\sigma_1, \sigma_2, R\sigma_2, \dots$ with independent σ_i), GATE, or STRICT (Section II). We write $r = 2m + \epsilon$ with $\epsilon \in \{0, 1\}$.

IV. GUARANTEES AND LIMITS

A. When is pairing safe?

Theorem 1 (Safety of independent reversal pairs). *Under the legal-order distribution of Section III, $B \leq f^2$ in each of the following cases:*

- Arbitrary nested groupings with a common cleaner set, disjoint from the polluters; or a brittle with at least one state-setter (then $B = 0$).*
- Unconstrained orders with arbitrary polluter-specific cleaners disjoint from the polluters.*
- Class-contiguous orders with disjoint roles, where no class that contains a polluter, the victim’s class included, contains a cleaner of any polluter outside that class.*
- Class-contiguous orders with disjoint roles, where every non-global cleaner of a polluter lies in that polluter’s class; a global cleaner cleans every polluter.*

For each such target, independent reversal pairs have detection probability at least that of independent random orders at every budget, both with and without a passing reference. Their expected number of detected targets is also no smaller.

Proof ideas; complete proofs in the supplement. Scan outward from v on each side; a side *fails* if it meets an active polluter. Case (a) is immediate for brittles, since a state-setter lies on one side of v .

(a) *A composable certificate.* For a subtree $T \ni v$, let u_T be the probability that T has no relevant test left of v ,

and t_T the probability of this together with right failure. Put $G_T(x) = (f_T + u_T x)^2 - B_T - 2t_T x$. At the lowest node with $k \geq 2$ relevant children, let the $k - 1$ siblings of v 's child have probabilities α_i that their first relevant test, seen from an end, is a polluter; let $S = \sum \alpha_i$ and $Q = \sum \alpha_i^2$. Uniform permutations give $f = S/k$, $B = (S^2 - Q)/[k(k-1)]$, $u = 1/k$, and $t = S/[k(k-1)]$, so

$$G(x) = \frac{1}{k^2} \left(x - \frac{S}{k-1} \right)^2 + \frac{Q - S^2/(k-1)}{k(k-1)} \geq 0.$$

The second term is nonnegative by Cauchy-Schwarz. At an ancestor with k relevant children whose other children have α -sum S , a sibling can only decide a side that T leaves open. Hence, with $h = S/k$, we get $f' = f + uh$, $B' = B + 2th$, $u' = u/k$, and $t' = t/k$, that is, $G_{\text{parent}}(x) = G_T(h + x/k)$. Nonnegativity on $\mathbb{R}$ survives this affine substitution, and $x = 0$ at the root gives $f^2 - B \geq 0$.

(b) *Cyclic cut*. Write $\sigma = AvC$ and map it to $T\sigma = CvA$, a measure-preserving bijection that commutes with reversal. In σ , a polluter $p \in A$ is active iff every cleaner of p in A precedes p . In $T\sigma$, this means that p comes after v and after all of its cleaners. With i.i.d. uniform heights Z , $F(\sigma)$ therefore becomes $M_+ = \{\exists p : Z_p > Z_t \ \forall t \in \{v\} \cup C_p\}$, and $F(R\sigma)$ becomes M_- , the same event with “ $<$ ”. With disjoint roles, M_+ increases in the polluters’ heights and decreases in all others, and M_- does the opposite. Harris’ inequality [9] gives $B = \mathbb{P}(M_+ \cap M_-) \leq f^2$.

(c) *Conditional monotonicity*. Use i.i.d. uniform class and method times, and condition on the victim’s times (u, w) . Order all other coordinates cyclically away from the victim. Then left failure is increasing and right failure is decreasing, so Harris’ inequality gives $B \leq \iint a(u, w) a(1-u, 1-w)$, where a is the conditional failure probability. One-sidedness makes a monotone in each coordinate, and Harris’ inequality on $[0, 1]^2$ bounds the integral by f^2 .

(d) *Window identity and composition*. Classify each side by its first decisive event: fail, clean (a global cleaner), or undecided. For the victim’s class alone, let u_0 , t_0 , z_0 be the probabilities that the left side is undecided, that it is undecided while the right side fails, and that both sides are undecided. Put $H_0(x) = (f_0 + u_0 x)^2 - B_0 - 2t_0 x - z_0 x^2$. Condition on the positions of v and the g global cleaners in its class. The windows they cut are adjacent uniform spacings whose law does not depend on v ’s rank among them. For $g \geq 1$, this gives $u_0 = (1 - f_0)/(g + 1)$, $t_0 = (f_0 - B_0)/(g + 1)$, and $z_0 = 0$, hence

$$H_0(x) = d_0 \left(1 - \frac{x}{g+1} \right)^2 + q_0 \left(\frac{x}{g+1} \right)^2 \geq 0 \quad \text{for all } x \in \mathbb{R},$$

with $d_0 = f_0^2 - B_0 \geq 0$ by case (b). For $g = 0$, $H_0(x) = d_0(1 - x)^2$. Because non-global cleaners are class-local, the other classes form an independent ensemble of fail/clean/transparent items with its own $H_E \geq 0$. Resolving undecided sides by the ensemble gives the exact identity $H(x) = H_0(h + ax) + z_0 H_E(x)$, where h and a are

TABLE II
SCOPE OF PAIRING SAFETY ($B \leq f^2$). LETTERS REFER TO THEOREM 1.
“OPEN” IS NOT USED BY THE EMPIRICAL CONCLUSIONS.

Order structure	Proven scope and boundary
Unconstrained	Common or arbitrary specific cleaners with disjoint roles: (a), (b).
Class-contiguous	Common cleaners: (a); specific cleaners satisfying (c) or (d). General disjoint case open; overlapping roles can fail (7 tests).
Deeper nesting	Common cleaners: (a). Specific cleaners can fail (6 tests; Example 2).

the ensemble’s failure and transparency probabilities, and $f^2 - B = H(0) \geq 0$. $\square$

The invariant $H \geq 0$ on all of $\mathbb{R}$ says that every scoring of the three side states has nonpositive left-right covariance. Because it composes, it also lifts along the victim’s path in deeper hierarchies (supplement).

What safety means. For $0 < f < 1$, the conditional chance that a passing order’s reversal fails is $(f - B)/(1 - f) \geq f$. An independent pair misses with probability $q = 1 - 2f + B \leq (1 - f)^2$. Multiplying over independent pairs proves detection dominance; linearity gives the expected-count result without assuming independence between targets. The guarantee is per target: it does not cover the time to find every OD test, the chance of finding at least one, or a suite-level gate.

The open class-contiguous case has neither condition (c) nor (d). Exhaustive search found no counterexample among 578,658 instances with at most seven tests; this is not a proof. All dataset models already satisfy proved cases, so resolving the conjecture is not needed for our conclusions.

B. A map of what scheduling can buy

Computing detection at a budget. For $r = 2m + \epsilon \geq 1$ runs,

$$\begin{aligned} D_{\text{PAIR}}^{\text{KR}} &= 1 - q^m(1 - f)^\epsilon, & D_{\text{IID}}^{\text{KR}} &= 1 - (1 - f)^r, \\ D_{\text{PAIR}}^{\text{NR}} &= D_{\text{PAIR}}^{\text{KR}} - B^m f^\epsilon, & D_{\text{IID}}^{\text{NR}} &= D_{\text{IID}}^{\text{KR}} - f^r. \end{aligned} \quad (2)$$

Without a reference, subtract the probability that every run fails, since observing both outcomes is required.

Theorem 2 (Block limit). *Let a schedule consist of independent blocks of k runs. Each run is uniformly distributed, and dependence within a block is arbitrary. Let T be the number of runs until a target’s first failure, with $0 < f < 1$. If $kf \leq 1$, then*

$$\mathbb{E}T \geq \frac{1}{f} - \frac{k-1}{2},$$

with equality iff the failure events within each reached block are pairwise disjoint. For reversal pairs ($k = 2$), $\mathbb{E}T_{\text{PAIR}} = (2 - f)/(2f - B)$ exactly, and

$$0 \leq \mathbb{E}T_{\text{IID}} - \mathbb{E}T_{\text{PAIR}} = \frac{f^2 - B}{f(2f - B)} \leq \frac{1}{2}$$

when $B \leq f^2$, with equality iff $B = 0$.

Proof. Let $d_i \leq f$ be the probability that a block's first failure is its $(i+1)$ -st run, and let $h = \sum_i d_i$. For identically distributed blocks, summing survival probabilities gives $\mathbb{E}T = [k(1-h) + \sum_i (i+1)d_i]/h$. For $h = \rho f$, the sum $\sum_i (i+1)d_i$ is smallest when mass f sits on the earliest runs, so it is at least $f\rho(\rho+1)/2$. Hence $\mathbb{E}T \geq k/(\rho f) - k + (\rho+1)/2$. This is decreasing in $\rho \in [0, k]$ because $\rho^2 \leq k^2 \leq k/f$, and $\rho = k$ gives the bound. Equality forces $h = kf$ and $d_i = f$ for all i , i.e., disjoint failure events. The supplement extends the bound to independent blocks with different laws by weighting each block's expected cost by its reach probability. For pairs, $\mathbb{P}(T > 2j) = q^j$ and $\mathbb{P}(T > 2j+1) = q^j(1-f)$, which gives $(2-f)/(1-q)$; the bound $\leq \frac{1}{2}$ rearranges to $B(2-f) \geq 0$. $\square$

In the $kf \leq 1$ regime, independent blocks of bounded size thus buy at most $(k-1)/2$ runs. A test with $f = 0.01$ needs at least 99.5 runs on average under PAIR, not 50. Within this block family, a constant-factor reduction requires blocks of order $1/f$. Information-directed nonuniform scheduling is a different way to leave the family; the next theorem gives two benchmarks for such schedules.

Theorem 3 (Lower bounds beyond independent blocks). *For a target with $0 < f < 1$:*

- (a) *Any schedule whose runs are individually uniform, with arbitrary dependence, satisfies*

$$\mathbb{E}T \geq L(f) := (N+1)(1 - Nf/2), \quad N = \lfloor 1/f \rfloor.$$

- (b) *Any policy issuing i.i.d. fresh orders in sequence or reversals of already issued fresh orders satisfies*

$$\mathbb{E}T \geq O(f, B) := \frac{1 + f - B}{2f - B}.$$

This includes adaptive policies and policies with an oracle. An oracle that reverses a passing fresh order exactly when its reversal fails attains (b) for that target.

Proof ideas. For (a), the union bound gives $\mathbb{P}(T > t) \geq (1 - tf)^+$; summing yields $L(f)$. For (b), let i^* be the first fresh order whose reversal orbit contains a failure. It is geometric with parameter $2f - B$. Every policy must issue at least i^* fresh orders, plus one run if that first useful order passes, giving $[1 + (f - B)]/(2f - B)$. The supplement proves both results, including target-aware attainment of (a). $\square$

Interpretation: a benchmark, not a delivered speedup.

For rare tests with $B = 0$, both bounds are about $(1/f + 1)/2$, far below $1/f - 1/2$ for pairing. But these are *per-target* benchmarks: a target-aware optimum need not be achievable by a single schedule serving all targets. Their average bounds the average detection time from below. Moreover, (a) and (b) constrain different classes: gates need not have uniform marginals, and arbitrary nonuniform failure-directed schedules need satisfy neither bound.

Without a passing reference. Let $T^\pm$ count runs until both outcomes occur. The supplement proves

$$\mathbb{E}T_{\text{PAIR}}^\pm = \frac{2-f}{2f-B} + \frac{1+f}{1-B} - 1, \quad \mathbb{E}T_{\text{IID}}^\pm = \frac{1}{f} + \frac{1}{1-f} - 1. \quad (3)$$

When $B \leq f^2$, pairing saves between zero and one run; one run is attained only at $f = 1/2$, $B = 0$. Apply the half-run bound separately to the first failure and the first pass.

Rare failures gain little at any budget. Under KR, $\Delta_r = D_{\text{PAIR}} - D_{\text{IID}}$ obeys

$$0 \leq \Delta_r \leq \lfloor r/2 \rfloor (f^2 - B)(1-f)^{r-2}.$$

For $0 < f \leq 1/2$, its budget-uniform upper envelope is $f^2/[-2e(1-f)^2 \ln(1-f)] = f/(2e) + O(f^2)$. The leading term is asymptotic, not an exact finite- f upper bound. This explains why a large conditional probability improvement can produce a small absolute detection gain.

C. What does not transfer to the deployed tool

Proposition 1 (Gated reversal changes the order law). *Let $\sigma \sim \mu$. With probability $g(\sigma)$ run $R\sigma$ next, and otherwise run a fresh $\eta \sim \mu$. The law ν of the second run satisfies $\frac{d\nu}{d\mu}(\tau) = 1 + g(R\tau) - \mathbb{E}_\mu g$. Hence $\nu = \mu$ iff g is almost surely constant. A deterministic gate that accepts a set of mass a moves the law by total variation $a(1-a)$.*

Proof. The reversal branch contributes $\int g(\sigma) \mathbf{1}\{R\sigma \in A\} d\mu = \int_A g(R\tau) d\mu(\tau)$, and the fresh branch contributes $(1 - \mathbb{E}g)\mu(A)$. $\square$

The union bound $D \leq \min(1, rf)$ holds for any schedule whose runs are individually uniform, but it cannot be used for gated schedules without their actual marginals. The same applies to per-target arguments. GATE conditions on *other* tests' outcomes, not on the target's own pass. Two targets with identical (f, B) can therefore behave differently under the gate, and no function of per-target statistics determines a gated suite's outcome.

Example 1 (The gate can lose to random). Two bits A, B start clear. Classes $\{v_A, p_B\}$, $\{v_B, p_A\}$, and $\{c\}$ contain victims that check their bit, polluters that set the other class's bit, and a cleaner c that clears both. Each victim satisfies the common-cleaner model. Of the 24 legal orders, 8 pass both victims; this set is closed under reversal. The remaining 16 orders fail exactly one victim, and reversal swaps which one. With two runs under KR, the expected numbers of detected victims are 10/9 for IID, 8/9 for the initial GATE step, and 4/3 for PAIR. For *at least one* detection, however, IID (8/9) beats PAIR (2/3).

Neither rule dominates the other. In the module with classes $\{v_0, v_1, c\}$ and $\{p\}$, where p pollutes both victims and c cleans only for v_1 , the gate detects 25/16 victims in two runs and pairing 3/2.

Example 2 (Three levels break the guarantee). For the tree $((v, (p_1, c_1)), (p_2, c_2), p_3)$ with $C_{p_1} = C_{p_2} = \{c_1\}$ and

$C_{p_3} = \{c_2\}$, all 32 legal orders give $f = 11/16$ and $B = 1/2$, so $B - f^2 = 7/256 > 0$. With two runs, pairing detects with probability $7/8$ and IID with $231/256$. The safety theorem therefore does not extend to nested groupings with polluter-specific cleaners (Table II). The limits of Section IV-B that do not use $B \leq f^2$ remain valid.

What a useful predictor must achieve. The oracle benchmark assumes perfect information. To make it actionable, consider a fixed predictor for one target: after each passing fresh order, it either requests its reversal or skips to a fresh order. Fresh-order/predictor cycles are independent. Let s be its true-positive rate on pass/fail pairs and a its false-positive rate on pass/pass pairs. Put $J = f - B$ and $q = 1 - 2f + B$. A renewal calculation gives

$$\mathbb{E}T_{s,a} = \frac{1 + sJ + aq}{f + sJ}, \quad (4)$$

$$\mathbb{E}T_{s,a} < 1/f \iff sJ(1 - f) > afq.$$

A cycle costs one fresh run plus a reversal with probability $sJ + aq$, and succeeds with probability $f + sJ$. This proves the formula. The full derivation, edge cases, and exact checks are in the supplement.

For a hypothetical target with $f = 0.01$, $B = 0$, a predictor with $s = 0.8$ and $a = 0.1$ needs 61.44 expected runs, versus 100 for random sampling, 99.5 for pairing, and 50.5 for the oracle. This is a calculated design point, not an evaluated predictor. If prediction costs c suite-run equivalents per fresh order, replace the numerator by $1 + c + sJ + aq$; the break-even condition becomes $sJ(1 - f) > f(c + aq)$. This gives predictor research a measurable target; a suite-level predictor must also balance conflicting targets.

Verification scope. Complete proofs, the unresolved frontier, and auxiliary identities are in the supplement. Exact checks enumerate 14,791 common-cleaner hierarchy models and 731,630 legal orders, check closed forms and small counterexamples, and test the scheduling identities.

V. EMPIRICAL STUDY

We ask:

- **RQ1:** Do the recorded OD models satisfy the hypotheses of Theorem 1, and do they show nonpositive reversal covariance?
- **RQ2:** How much does pairing gain per OD test, and how much headroom do the limits leave?
- **RQ3:** How do GATE, STRICT, and PAIR compare with IID at suite level?
- **RQ4:** Are the conclusions robust to semantics, data inconsistencies, and aggregation?
- **RQ5:** Do matched-order predictions agree with real executions in two buildable modules?

A. Subjects and Semantics

We use the public dataset of Li et al. [8]: 289 known OD tests (231 victims and 58 brittles) in 47 modules of 27 open-source Java projects. The dataset gives polluters and

TABLE III
REFERENCE-CONSISTENCY SENSITIVITY AT 20 RUNS: MODELLED
DETECTION GAIN OVER IID (PP, MONTE CARLO 95% CI HALF-WIDTH).
“CONSISTENT” RESTRICTS SCORING, NOT THE SUITE-LEVEL GATE’S
STATE.

Population	n	PAIR-IID		GATE-IID	
		KR	NR	KR	NR
S1, All	289	0.19 ± 0.01	0.33 ± 0.02	0.18 ± 0.01	0.43 ± 0.02
S1, Consistent	181	0.29 ± 0.02	0.31 ± 0.02	0.28 ± 0.02	0.27 ± 0.02
S2, All	289	0.19 ± 0.02	0.33 ± 0.02	0.19 ± 0.02	0.43 ± 0.02
S2, Consistent	181	0.30 ± 0.03	0.32 ± 0.03	0.30 ± 0.03	0.28 ± 0.03

cleaners, or state-setters, for every test, together with each module’s recorded original order. Our primary semantics **S1** is exactly that of Li et al.’s reference simulator: Eq. (1), with the 21 cleaner *groups* (“ $a|b$ ”) split into individual cleaners. The sensitivity semantics **S2** lets a group clean only if all of its members run between polluter and victim. Orders follow iDFlakies’ default detector type: a uniform class order and uniform method orders.

Simulation and execution. The 47-module policy comparison simulates the published models; Section V-G checks the abstraction against real executions of two complete modules. The abstraction predicts that the recorded original order fails for 108 of 289 targets, despite the recorded passing outcome. Our primary analysis keeps the published population; a second view scores only the 181 S1 reference-consistent targets. KR supplies an external passing reference, even when the abstraction disagrees with it; NR needs no reference.

B. Computation and Statistics

Per-test statistics (RQ1, RQ2). Every brittle and every victim whose polluters share one cleaner set disjoint from the polluters (258 tests) has exact rational f and B from the two-level closed forms in the supplement. We cross-checked each closed form with 2×10^5 sampled orders; across 516 comparisons the largest $|z|$ was 2.73. For the other 31 victims we estimate f and B from 2×10^6 sampled orders each and propagate uncertainty by a parametric bootstrap of the four outcome cells. Per-test gains and expected run counts then follow from Eqs. (2)–(3) and Theorem 2.

Suite-level simulation (RQ3). We simulate all targets of a module jointly on the same orders, with 100,000 independent detector runs of 40 rounds per module and policy. GATE reproduces the tool’s rule: reverse iff the previous round detected no new target and the previous order was not itself a reversal; fresh orders are not de-duplicated. Under NR we adapt “new” to “newly detected under NR”. All four policies consume the same stream of fresh orders (common random numbers). We report the percentage of all 289 OD tests detected, obtained by summing expected detections over modules, with 95% confidence intervals. Per-module contrasts use a Bonferroni correction over 47 modules. Outcomes are deterministic for a given order. We model neither confirmation reruns, nor unknown OD or non-OD flaky tests, nor wall-clock cost.

TABLE IV
THEOREM COVERAGE, HYPOTHESIS AUDIT, AND FAILURE STATISTICS (S1).

OD tests (victims / brittles)	289 (231 / 58)
Covered by Thm. 1(a) / (c) / (d)	259 / 20 / 10
Exact closed form for f, B	258
victims with common cleaners / brittles	200 / 58
Estimated by Monte Carlo	31
with $d > 0$ supported (pointwise 95% CI)	31
with $d < 0$ supported (pointwise 95% CI)	0
Exact tests with $d = 0$ (no strict gain)	0
Tests with $B = 0$ (attain half-run limit)	211
Flake rate f : mean / median / min	41.7 / 50.0 / 1.2 %
Tests with $f < 0.1$	71
Pass the recorded original order under the model	181

Implementation validation, not execution validation. (i) For IID and PAIR, the expected number of detected targets equals the sum of per-target closed forms by linearity. Across 6,272 module/budget/protocol comparisons, the largest $|z|$ was 3.26. (ii) We ran the *unmodified* simulator of Li et al. on 150 of our sampled orders per module. It reports the same first passing and first failing order for all 289 tests.

C. RQ1: Hypotheses and Covariance

Table IV summarises the audit. For the class-contiguous orders that iDFlakies samples, proven theorems cover all 289 tests. Theorem 1(a) covers 259: all brittles, 200 victims whose polluters share their cleaners, and one victim whose only polluter is listed as its own cleaner (a vacuous entry, which the reference simulator also ignores). Theorem 1(c) covers 20 victims whose classes are one-sided. Theorem 1(d) covers the 10 `activiti` victims, whose polluters share one global clean-up test per class and have a few class-local cleaners. For the 31 tests without a two-level closed form, the pointwise 95% intervals also support $d > 0$. No test in the dataset shows positive covariance, so Example 2 remains a theoretical boundary here. No exact test has $d = 0$, so the conditional-probability effect of Theorem 1 is strict for all of them. 211 tests have $B = 0$: no legal order fails in both directions, usually because the victim has one polluter or the test is a brittle.

Answer: The safety theorem covers all 289 recorded OD models under the orders iDFlakies uses. All 289 models have nonpositive reversal covariance, so independent pairing never lowers their detection probability.

D. RQ2: Per-Test Benefit Versus the Limits

Fig. 1(a) plots the expected saving until the first failure. The 211 tests with $B = 0$ save exactly half a run, as Theorem 2 predicts. The mean saving is 0.40 runs: 10.22 runs under IID versus 9.82 under PAIR, a relative saving of 12.5% on average. Without a reference the mean saving is 0.70 runs (12.62 versus 11.91), and 94 tests attain the one-run limit of Eq. (3). Reversal thus already delivers nearly all of the benefit that the theory allows, but that benefit is at most a fraction of one run.

TABLE V
PER-TEST DETECTION (%) AND GAIN OF PAIR OVER IID, AVERAGED OVER 289 OD TESTS (S1). BOOTSTRAP 95% CIs ARE NARROWER THAN ± 0.01 PP.

Budget r (runs)	2	4	10	20	40
KR IID	58.0	71.2	81.2	86.9	92.8
KR Pair	69.1	74.8	81.7	87.1	92.9
KR mean gain (pp)	11.18	3.51	0.46	0.19	0.13
KR bound $mf^2(1-f)^{r-2}$ (pp)	25.46	7.03	0.72	0.21	0.13
KR max gain, one test (pp)	25.00	8.64	2.99	1.42	0.69
NR IID	32.5	58.0	75.5	84.0	91.7
NR Pair	54.9	64.9	76.4	84.4	91.9
NR mean gain (pp)	22.37	6.91	0.85	0.33	0.18
NR max gain, one test (pp)	50.00	12.50	2.99	1.42	0.69

TABLE VI
WHERE THE BENEFIT AND THE MISSES ARE (KR, S1): MEAN FLAKE RATE, SHARE OF TESTS WITH $B = 0$, MEAN RUNS SAVED, MEAN GAIN AT $r = 10$, AND SHARE OF ALL IID MISSES AT $r = 20$.

Group	n	$\bar{f}$ %	$B=0$ %	saved	gain	misses %
Brittles	58	10.8	100	0.50	0.55	67
Victims, one polluter	123	38.2	100	0.50	0.55	23
Victims, ≥ 2 polluters, common cleaners	78	70.6	12	0.14	0.19	7
Victims, polluter-specific cleaners	31	41.2	71	0.42	0.60	4

Table V shows the consequences for fixed budgets. Pairing adds 11.18 pp at $r = 2$, but only 0.46 pp at $r = 10$ and 0.19 pp at $r = 20$ under KR. Under NR the corresponding numbers are 22.37, 0.85, and 0.33 pp. The largest single-test gain at $r = 10$ is 2.99 pp. Fig. 1(b) shows that the gain follows the bound of the rare-failure bound in Section IV closely: it peaks for f around $1/r$ and vanishes for frequently failing tests, which are detected anyway. The tests that remain undetected are the rare ones. At $r = 10$ and $r = 20$, 94% and 98% of the IID misses (KR) come from the 71 tests with $f < 0.1$. For these tests the budget-uniform gain is $O(f)$; the exact envelope, rather than just its leading term $f/(2e)$, is given in Section IV.

Table VI explains why. Brittles and single-polluter victims always have $B = 0$ and save exactly half a run. Brittles, however, are rarely failing ($\bar{f} = 10.8\%$): a brittle fails only when it precedes all of its state-setters. Together, brittles and single-polluter victims account for 90% of the IID misses at $r = 20$. Victims with several polluters fail often and gain little. The budget needed for a given coverage barely changes: to detect 90% of the known OD tests, IID needs 29 runs and PAIR 29 under KR (34 and 34 under NR); for 95%, the numbers are 54 versus 53 (KR) and 58 versus 57 (NR).

Table VII places these numbers on the map of Section IV. The gap from IID to the uniform-marginal lower bound is 45% of its mean expected runs, and pairing closes 8.6% of that gap. Because per-target optima may conflict, this gap is an upper limit on what any detector could save, not a saving on offer. Four-run blocks remain limited, especially for rare tests. For those tests, reducing expected runs by a fixed fraction $\eta > 0$ through independent blocks necessarily requires $k \geq 1 + 2\eta/f$ in the $kf \leq 1$ regime.

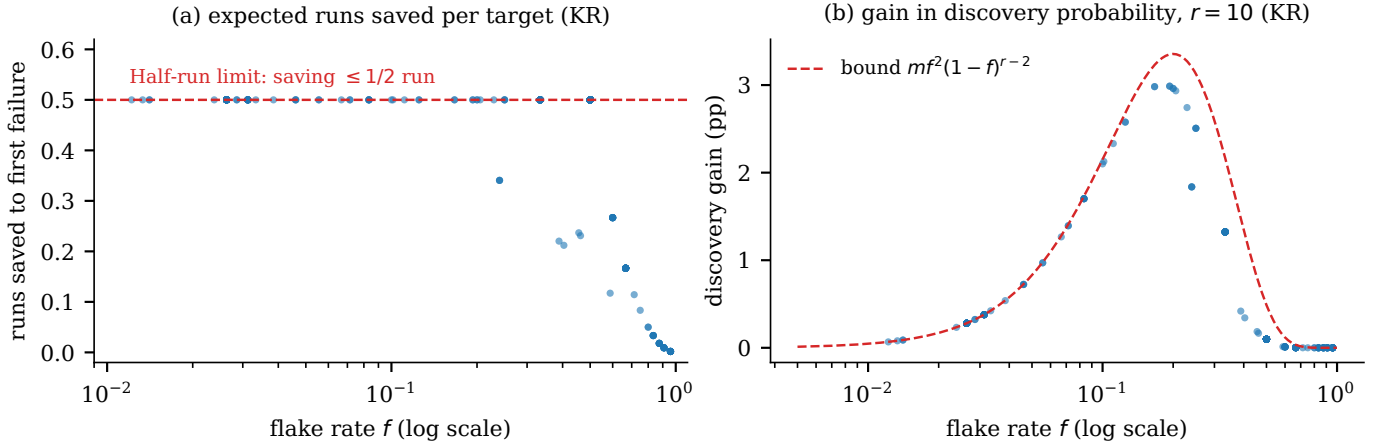


Fig. 1. Per-test benefit of independent reversal pairs versus flake rate (KR, S1). (a) Expected runs saved until the first failure: 211 tests sit exactly on the half-run limit of Theorem 2. (b) Gain in detection probability after 10 runs against the rare-failure bound in Section IV; most tests lie on the bound because $B \approx 0$.

TABLE VII

THE LIMITS MAP ON REAL DATA: EXPECTED RUNS TO THE FIRST FAILURE (KR), AVERAGED OVER ALL 289 OD TESTS AND OVER THE 71 RARE ONES ($f < 0.1$).

Schedule (theorem)	all	rare
IID ($1/f$)	10.22	34.66
PAIR, outcome-blind (Thm. 2, exact)	9.82	34.16
4-run blocks (lower bound)	9.45	33.16
reverse-or-fresh (per-target oracle)	5.62	17.83
uniform marginals (lower bound)	5.62	17.83

TABLE VIII

SUITE-LEVEL DETECTION (% OF 289 OD TESTS) AND DIFFERENCES IN PERCENTAGE POINTS WITH 95% CI HALF-WIDTHS (S1).

Budget r (runs)	2	4	10	20	40
<i>No reference (ISSTA'23 definition)</i>					
IID	32.5	57.9	75.5	84.0	91.7
Pair	54.9	64.9	76.4	84.4	91.9
Gate (iDFlakies)	54.9	64.4	76.4	84.5	92.0
Strict gate	34.8	59.2	75.8	84.2	91.8
Δ Pair-IID	$+22.40 \pm 0.04$	$+6.94 \pm 0.03$	$+0.85 \pm 0.02$	$+0.33 \pm 0.02$	$+0.18 \pm 0.01$
Δ Gate-IID	$+22.40 \pm 0.04$	$+6.48 \pm 0.03$	$+0.86 \pm 0.02$	$+0.43 \pm 0.02$	$+0.24 \pm 0.01$
Δ Pair-Gate	$+0.00 \pm 0.00$	$+0.46 \pm 0.01$	-0.01 ± 0.00	-0.10 ± 0.00	-0.06 ± 0.01
<i>Known passing reference</i>					
IID	58.0	71.2	81.2	86.9	92.8
Pair	69.1	74.8	81.7	87.1	92.9
Gate (iDFlakies)	60.3	74.4	81.7	87.1	92.9
Strict gate	60.3	72.5	81.5	87.1	92.8
Δ Pair-IID	$+11.19 \pm 0.04$	$+3.54 \pm 0.02$	$+0.46 \pm 0.01$	$+0.19 \pm 0.01$	$+0.13 \pm 0.01$
Δ Gate-IID	$+2.31 \pm 0.02$	$+3.14 \pm 0.02$	$+0.43 \pm 0.01$	$+0.18 \pm 0.01$	$+0.12 \pm 0.01$
Δ Pair-Gate	$+8.88 \pm 0.03$	$+0.40 \pm 0.01$	$+0.02 \pm 0.00$	$+0.00 \pm 0.00$	$+0.00 \pm 0.00$

Answer: Pairing saves half a run for 211 tests and 0.40 runs on average under the model. Its aggregate detection gain is below 1 pp from 10 runs on. Closing the remaining gap requires longer-horizon or information-directed designs.

E. RQ3: Suite-Level Policies

Table VIII and Fig. 2 compare the four policies on complete modules. Three observations stand out.

First, the differences are real but confined to tiny budgets. Under KR, PAIR improves on IID by +11.19 pp at $r = 2$ and by +3.54 pp at $r = 4$, but only by +0.46 pp at $r = 10$ and +0.19 pp at $r = 20$. The picture under NR is the same: +22.40, +6.94, +0.85, and +0.33 pp.

Second, we observe no significant loss for the modelled gate relative to random sampling. GATE never falls significantly below IID in any module, under either protocol and at any budget from 1 to 40 runs; the most negative module-level z -score is -1.0 , even before correcting for multiple comparisons. This is absence of detected harm rather than an equivalence test. From 10 runs on, GATE and PAIR differ by at most 0.11 pp in aggregate: +0.02 and +0.00 pp under KR, and -0.01 and -0.10 pp under NR. Per module the sign varies: at $r = 10$ under NR, PAIR is significantly better in 14 modules and worse in 5. Under NR the gate reverses after the first round almost always, because one run cannot reveal both outcomes, so it behaves like PAIR early on.

Third, the gate matters only under a tiny budget with a passing reference. There the first round often detects something new, so GATE forgoes the reversal and falls behind PAIR by +8.88 pp at $r = 2$. STRICT is weaker still at small budgets, because a run without any failure is rare in modules with many OD tests.

Table IX shows the largest modules. Where detection is already high, reversal adds about 1 pp or less at $r = 10$ (e.g., +1.3 pp for `lib` under KR). Where it is low, reversal adds less than 0.5 pp. In `dubbo-config-api` (23%), 37 brittles fail in only 2.6% of orders. In `marine-api` (27%), 12 victims fail in 3.1% of orders. The largest per-module effects (+3.0 pp) appear under NR in modules whose OD tests fail in most orders, e.g., `integration-test-3_7` ($f = 0.83$). There, detection needs a *passing* run, the mirror image of the rare-

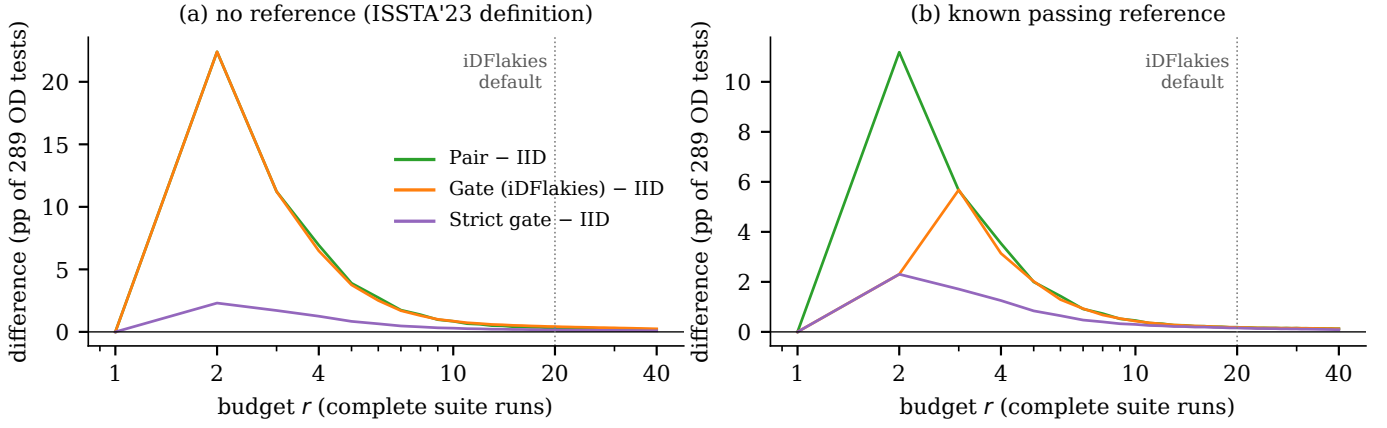


Fig. 2. Suite-level difference to IID in detected OD tests (percentage points of 289), with 95% confidence bands (S1, 100,000 detector runs per module). The dotted line marks iDFlakies’ default budget of 20 runs.

TABLE IX
DETECTION (%) AFTER 10 RUNS IN THE 13 MODULES WITH AT LEAST FIVE OD TESTS (227 OF 289 TESTS; S1).

Module	OD	KR			NR		
		IID	PAIR	GATE	IID	PAIR	GATE
naming	44	98.9	99.1	99.1	98.9	99.1	99.1
dubbo-config-api	37	23.4	23.7	23.6	23.4	23.7	23.5
hadoop-hdfs-nfs	28	100.0	100.0	100.0	98.9	99.7	99.7
lib	25	98.3	99.6	99.5	98.3	99.6	99.3
activiti-spring-boot-s...	16	95.0	96.1	95.9	94.9	96.1	95.8
spring-ws-core	15	100.0	100.0	100.0	34.7	35.3	37.4
wro4j-core	15	92.4	92.6	92.6	92.3	92.6	92.6
marine-api	12	27.2	27.6	27.6	27.2	27.6	27.7
integration-test-2_1	9	99.4	99.6	99.6	65.1	67.0	66.6
spring-boot-actuator-a...	8	99.9	100.0	100.0	99.8	100.0	100.0
integration-test-2_2	7	100.0	100.0	100.0	73.6	76.2	75.6
integration-test-3_7	6	100.0	100.0	100.0	83.8	86.8	86.2
hadoop-mapreduce-clien...	5	100.0	100.0	100.0	99.2	99.9	99.9

failure case. In `spring-ws-core` ($f = 0.96$, NR) the gate even beats unconditional pairing, by 2.1 pp.

Answer: In aggregate under S1, all modelled reversal policies are within 1 pp of random orders from 10 runs on. The deployed gate is never significantly worse than random and is within 0.11 pp of unconditional pairing at realistic budgets. Its clearest aggregate advantage over the modelled gate occurs with a passing reference and two to four runs; individual modules can differ.

F. RQ4: Robustness

Semantics. Under S2 (conjunctive cleaner groups), 254 tests are exact and all 35 estimated tests again have $d > 0$. The suite contrasts change by at most 0.04 pp at any budget; e.g., PAIR–IID at $r = 10$ is +0.46 pp (KR) and +0.86 pp (NR).

Reference-order consistency. Table III reports both populations at the default budget without hiding the inconsistent targets. The consistent view retains the original gate decisions, including decisions influenced by inconsistent targets; it is not a rerun on a cleaned suite. The abstraction predicts a *failure* of

the recorded original order for 108 tests, despite the historical passing-reference assumption. For 59 of the 104 affected victims, every uncleaned polluter lies in another class; for the other 45, an uncleaned polluter shares the victim’s class. Class-level resets or omitted cleaners are possible explanations. The six such targets in our executed `http-request` module really fail in the recorded order (Section V-G), so reference inconsistency alone does not indicate model error. Restricted to the 181 consistent tests, PAIR–IID at $r = 10$ is +0.62 pp (KR) and +0.86 pp (NR), and GATE–IID is +0.59 pp (KR).

Aggregation. Weighting modules equally instead of tests gives PAIR–IID at $r = 10$ of +0.73 pp (KR) and +1.21 pp (NR).

Answer: The small pairing gains persist under alternative cleaner-group semantics, scoring only reference-consistent targets, and equal-module weighting. RQ5 addresses execution fidelity.

G. RQ5: Bounded Real-Execution Check

Frozen selection and protocol. Before building, we fixed two primary modules and two technical-failure fallbacks, without selecting on observed model agreement. The primaries were `aismessages` (two reference-consistent OD targets, model $f = 2/3$, $B = 1/3$) and `http-request/lib` (25 targets, six reference-inconsistent, model $f = 1/3$, $B = 0$). Both built at the dataset revisions without modifying production code, tests, dependencies, or fixtures; no fallback was used. Native Maven/JUnit discovery exactly matched the recorded inventories of 44 and 163 tests, respectively. This is a feasibility-selected case study, not a representative sample.

Each execution ran the complete module in a fresh JVM with normal class and method fixtures. The ordered JUnit runner retained whole-class execution and checked every actual test start/finish against the requested sequence. Three repetitions of the recorded order and two repetitions of five fixed random orders gave stable outcomes in both modules; these 26 pilot runs were excluded. The formal sample was frozen at 300 independent uniform class-contiguous orders per

TABLE X

REAL EXECUTIONS: COMPLETE-SUITE SIZE / OD TARGETS, INDEPENDENT PAIRS, RANGES OF OBSERVED ANCHOR $\hat{f}$ AND JOINT B ACROSS TARGETS, AND S1 MISMATCHES / TARGET OUTCOMES. THESE RANGES ARE NOT CONFIDENCE INTERVALS.

Module	tests/OD	pairs	$\hat{f}$	$\hat{B}$	mismatch
aismessages	44/2	300	0.673	0.293–0.307	0/1200
http-request	163/25	300	0.313–0.407	0.000	0/15000

module, each followed by its whole-order reversal in a separate fresh JVM. All 1,200 formal executions completed with exact order agreement and no skips or timeouts. S1 and S2 predicted outcomes on these same orders, without model adjustment. Tool versions, regression tests, and per-target intervals are in the supplement and package.

Results and uncertainty. All 16,200 formal target outcomes matched S1 and S2 (Table X). Agreement is order by order, which is stronger than matching marginal rates. With the pair as the independent unit, zero pairs containing any target mismatch gives a one-sided 95% upper bound of $1 - 0.05^{1/300} \approx 0.00994$ on that event’s probability, separately for each module. Zero observed joint failures in `http-request` are likewise consistent with, not proof of, $B = 0$. We report pointwise Wilson intervals using 300 pairs, not 600 independent directions. For 2 of the 27 targets, the model’s theoretical f lies outside the anchor-rate 95% interval; the model on the *same sampled orders* has the identical empirical rate, so these deviations are not execution/prediction disagreements.

The reference-order discrepancy is observable. Native Maven baselines passed all tests, but executing the recorded original order made six known `http-request` targets fail, consistently with S1/S2, in all three repetitions. For these targets the model is right and the historical passing reference does not reproduce. Three tests outside the published target set also failed during sampling; we report them but did not add them to the targets.

Answer: The abstraction predicts every sampled outcome of these 27 targets exactly, including the six reference-inconsistent ones. The check covers two modules with $f \geq 1/3$ (Section VII).

VI. DISCUSSION

Decision 1: do not sell a local scheduling change as a rare-test solution. Independent reversal pairs cannot save more than half a run per target. This result needs only equal uniform marginals, not a faithful polluter/cleaner abstraction. Generalising the transformation to independent fixed-size blocks still saves only a bounded number of runs in the rare-event regime. These limits are a stopping criterion for optimisation; reversal still helps at a two-run budget.

Decision 2: do not replace the deployed gate on these results alone. At 20 runs, our modelled aggregate differences are small. With a two-run KR budget, pairing exceeds the gate by +8.88 pp, which warrants evaluating unconditional pairing

for small per-commit budgets. The gate has no per-target safety guarantee, and confirmation reruns, unknown flaky tests, and historical-order suppression can change its behaviour.

Decision 3: evaluate information, not just reversal frequency. Shared-state analysis [10]–[12] and prioritisation [13] might supply information about useful reversals. Equation (4) states exactly how recall, false positives, and prediction cost affect expected runs for a fixed target, and so tells such tools which operating point they must reach. Longer-horizon covering designs [6], [8] are the other way out of the block limit.

Next steps. A tool-level evaluation that executes the competing policies should include rare OD tests, confirmation reruns, unknown flaky tests, and elapsed time.

VII. THREATS TO VALIDITY

Construct. The 47-module policy outcomes come from the polluter/cleaner abstraction used by prior work [6], [8]. The separate two-module execution check supports only its sampled cases, not all projects or unobserved orders. The abstraction does not reproduce the passing original order for 108 tests. We therefore report results for the consistent subset as well. The dataset lists no polluter groups, so OD tests that need several polluters jointly are not represented. We also ignore non-OD flakiness and failures that alter later behaviour. We do not model wall-clock cost, confirmation reruns, or the tool’s reuse of historical orders.

Unproven cases. Every S1 model in the dataset is covered by a theorem; this is a claim about the recorded model, not the underlying execution. The open two-level conjecture matters only for suites with cleaners that are neither global nor class-local and that sit in classes with polluters.

Internal. The simulation is validated against closed forms and against the reference simulator of Li et al. Monte Carlo error is reported and is at least an order of magnitude smaller than the effects we interpret. The theorems have complete proofs. Our package checks their small-instance consequences by exhaustive enumeration on all small hierarchies (14,791 models, 731,630 orders) and exact rational checks of the budget and hitting-time identities.

External. The simulation covers 47 Maven modules and one tool at one commit. Execution covers two feasibility-selected modules on one Linux/JDK environment, with model failure rates at least $1/3$; it provides no direct evidence for rare failures or environment portability. Other frameworks group tests hierarchically too (e.g., nested describe blocks in Jest [5]), and Theorem 1 covers arbitrary nesting, but the empirical magnitudes may differ. The theory’s hypotheses (common cleaners, independent anchors) are explicit, and Examples 1–2 show what fails without them.

VIII. RELATED WORK

OD test detection. DTDetector introduced reversal, randomised, and bounded exhaustive detection [3]. iDFlakies randomises class-contiguous orders and classifies flaky tests [4].

Other work detects state pollution or test dependencies directly [10]–[12], repairs OD tests [14], or detects them incrementally [15]. Empirical studies cover Python [16] and JavaScript [5]; see Parry et al. [2] for a survey. Systematic approaches cover test pairs with Tuscan squares [6], [8]. Li et al. found that most generated orders are unnecessary and called for prioritisation [8], which Iqbal et al. pursue [13].

Probabilistic analysis of test orders. Wei et al. derived flake rates for special cases and proposed the reversal change [6]. Wang et al. count failing orders with Alloy [17]. We prove the reversal inequality in general hierarchies, bound its value, and separate per-test from suite-level claims.

Antithetic sampling. Pairing a sample with a reflected one is the classical antithetic technique [18], [19], and its use for estimating probabilities is well studied [20]. Those results reduce estimator variance, and they help only when the pair is negatively correlated. Theorem 1 establishes that sign for OD-test semantics, using a certificate for hierarchies and Harris’ correlation inequality [9] for unconstrained orders. The sign is not generic (Table II). Theorem 2 and Eq. (3) concern detection times rather than estimator variance.

IX. CONCLUSION

Reversal’s value is bounded, even when its intuition is correct: independent pairs save at most half a run per target’s first failure, and independent small blocks cannot deliver constant-factor savings for rare failures. Under explicit OD-model hypotheses, pairing is safe; a suite-level gate is a different policy and does not inherit that guarantee. Simulation of 289 published models finds gains below 1 pp from 10 runs on. Real executions of two modules match all 16,200 predicted outcomes of 27 targets. Our predictor tradeoff makes the next question concrete: can useful failure information be obtained accurately and cheaply enough to outperform outcome-blind sampling?

DATA AVAILABILITY

Submission draft: the anonymous artifact URL must be added before submission; no public availability is claimed here. It contains the complete proofs, exact verification scripts, the suite simulator, the analysis scripts that generate every number, table, and figure in this paper, and the processed dataset of Li et al. [8] with provenance.

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